Two-Timescale Resource Management for Ultrareliable and Low-Latency Vehicular Communications

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***Abstract*— Ultra-reliable low-latency communication (URLLC) is essential for future vehicle-to-vehicle (V2V) networks to improve traffic safety and enhance driving experience. Due to the fast-varying channel caused by high mobility, guaranteeing latency and reliability performance of the V2V links is a tremendous challenge. In this paper, we propose a novel resource allocation framework to support ultra-reliable low-latency V2V communications. The proposed framework includes both large-scale and small-scale resource optimizations. The large-scale resource allocation is performed at the central base station based on large-scale channel information periodically collected from vehicles. On the other hand, the small-scale resource allocation is performed at the vehicles according to instantaneous channel and queuing information. We develop optimal solutions for both resource allocation problems. With the proposed optimal solutions, the latency performance at the occurrence of extreme events is enhanced by enabling spectrum sharing among the vehicles. Simulation results demonstrate that the proposed algorithm can effectively improve the URLLC performance compared against the benchmark algorithm.**

***Index Terms*— 5G, ultra-reliable low-latency communications (URLLC), vehicle-to-vehicle (V2V) communications, resource allocation.**

I. Introduction

VEHICLE-TO-VEHICLE (V2V) communication, as an important application scenario in the 5th-generation (5G) network, is regarded as one of the most promising enablers for autonomous driving and intelligent transportation systems [1]–[3]. The communication link between vehicles supports various autonomous applications such as forward collision warning, lane change warning, and real-time navigation, which can significantly improve traffic safety and enhance driving experience. However, these safety applications are generally time-critical and reliability-sensitive, which require the network to provide very high quality-of-service (QoS). Therefore, achieving ultra-reliable low-latency communication (URLLC) is indispensable for V2V networks. To be specific, the METIS project requires end-to-end latency of less than 5 ms and transmission reliability of 99.999% for small packages [4]. In order to meet such critical latency and reliability requirements, the device-to-device (D2D) communication enabled cellular network has been proposed, which employs a central node to perform resource allocation and enables local information exchange among nearby devices [5]. This scenario is part of the present-day technology New Radio (NR) V2X, which is undergoing extensive developments to support advanced vehicular applications requiring high reliability, low latency, and high throughput communications [6], [7].

Achieving URLLC brings new challenges to the 5G cellular network. In general, the conventional communication systems are mainly developed to optimize the average performance, e.g., throughput, spectral efficiency, and average latency. For instance, the methods in [8]–[11] aim to provide good performance on average metrics in vehicular networks, which cannot support the strict URLLC requirements at the occurrence of extreme events. To overcome this challenge, the latency and reliability constraints need to be modeled properly and the occurrence of extreme events must be taken into consideration [2]. In [12], a probabilistic constraint has been proposed and imposed on the optimization problem to shorten the queue length of V2V pairs. In [13], the reliability and latency metrics of autonomous vehicular networks have been formulated and jointly optimized by utilizing the Euclidean norm theory. In [14], an integrated communication and control framework has been proposed to analyze the latency and reliability performance and select the control parameters of wireless connected vehicular platoons. Considering the mmWave signal propagation characteristics and the road topology, a performance analytical framework has been developed and a relay selection scheme has been proposed to improve the latency and reliability performance of mmWave multi-hop V2V communications [15]. In [16], the distribution of queue length is modeled by extreme value theory (EVT), which is a powerful tool to characterize the occurrence probability of extreme events [17]. Based on this, the sum power of V2V links is minimized while guaranteeing the queue stability.

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In order to reduce communication overhead, the work in [18] has proposed a Lyapunov-based distributed resource allocation algorithm to reduce the queuing latency by taking into account both EVT and federated learning. In [19], the authors have proposed a non-cooperative radio resource management algorithm for V2V networks based on the game theory, where V2V pairs compete for the limited frequency resource at the beginning of each scheduling slot with the aid of the RSU. In [20], the authors have employed the age of information (AoI) as the latency metric and modeled the tail distribution of AoI using EVT.

The work in [21] has proposed a URLLC-aware task offloading scheme which reduces the queuing latency of Internet of Health Things (IoHT). In [22], the coordinated multi-point (CoMP) transmission is utilized and a multi-timescale optimization problem is formulated to maximize the slice utilities of multicast enhanced mobile broadband (eMBB) and bursty URLLC service multiplexing. A two-phase transmission protocol for ultra-reliable industrial automation has been developed in [23], where messages are multicasted from the base station to different device groups and then relayed through D2D links to enlarge the signal coverage. The work in [24] has investigated the resource block (RB) sharing scheme for URLLC-oriented D2D communications. To minimize the collision-free probability, a RB selection game is modeled by game theory and the corresponding optimal strategy is developed. The deep learning technique has been investigated to guarantee the latency and reliability performance in communication systems [25], [26]. In [25], deep-RL has been utilized to perform risk-aware resource allocation by taking the latency and reliability statistics as the feedback. In [26], a generative adversarial network (GAN) has been developed to generate synthetic data and pre-train the deep reinforcement learning (DRL) network which can enhance the system robustness in practical environments. To learn the optimal control policy with dynamic channel and queue states, a proactive algorithm based on long short-term memory and DRL has been proposed in [27]. Leveraging the well-trained model and the observed network state, the RSU makes the optimal frequency resource allocation and packet scheduling decision at each scheduling slot. Due to the inherent inexplicability, the performance of deep-learning-based approaches is hard to be controlled in the URLLC systems, especially when the network is not fully trained.

Most of the above works require instantaneous channel state information (CSI) at the central node, which is difficult to be obtained in dynamic vehicular environments [12], [16], [20], [21], [25], [26], [28], [29]. To tackle this problem, the delayed CSI feedback of V2V links has been leveraged to optimize the spectrum and power allocation while guaranteeing the channel outage probability [30]. The method in [31] exploits the large-scale channel information and optimizes the ergodic capacity while guaranteeing the outage probability of V2V links. In [32], the reliability and latency requirements are transformed into optimization constraints according to the large-scale channel information. In [33], to avoid the frequent exchange of instantaneous CSI, the effective capacity has been introduced to model the URLLC performance of V2V links, where the data arrival rate is assumed to be constant. The work in [34] develops a multi-scale resource allocation method for vehicle-to-infrastructure (V2I) URLLC systems by utilizing large-scale CSI in the short-term optimization. In [35], the authors proposes a distributed federated learning method to model the extreme queue lengths with minimal coordination among V2V pairs. The queue length distribution model is averaged at the RSU side, while the instantaneous CSI and queue state are utilized by the V2V pairs to control the queue length.

Although the existing works have enhanced the URLLC performance of V2V networks from various aspects, the CSI and queue state exchange overhead still remains as an obstacle to effectively allocating radio resource for V2V pairs. Utilizing the instantaneous network state at the RSU would lead to better performances on latency and reliability, though the heavy information exchange overhead may cause the degradation on system performance [19], [20], [27]. On the other hand, the methods based on the large-scale CSI can significantly reduce the data exchange overhead [31]–[34]. However, the system performance may not be optimal since the instantaneous CSI is not utilized in the resource management algorithms. To reduce the data exchange overhead while utilizing the instantaneous CSI and queue state, some works have investigated the semi-distributed resource management, where each vehicle chooses the optimal control policy based on the local CSI and queue state and periodically communicates with the RSU to update the global queue length distribution model [18], [35]. However, these methods may not provide optimal resource management results, because the large-scale CSI is not utilized by the RSU and the spectrum resources are heuristically allocated to vehicles.

Inspired by this, we propose a novel resource management framework to support ultra-reliable low-latency V2V communication in D2D-enabled cellular systems. Our goal is to minimize the transmit power of V2V links while guaranteeing the strict latency and reliability constraints. To reduce the data exchange overhead, the large-scale resource allocation is performed at a central node based on the large-scale CSI of V2V pairs. By enabling spectrum sharing among different V2V pairs, the latency performance is enhanced at the occurrence of extreme events. Moreover, the instantaneous CSI and queue state are utilized to formulate the small-scale power allocation problem at V2V pairs to further guarantee the URLLC performance. In brief, the main contributions of this paper are summarized as follows.

- We propose a two time-scale resource allocation framework which combines large-scale and small-scale resource optimizations. In this framework, the large-scale CSI is transmitted from the vehicle pairs to the central node periodically. Leveraging the gathered CSI, the central node performs a large-scale resource optimization and transmits the resource allocation result to each vehicle pair, which continues to perform a small-scale resource optimization using the instantaneous CSI without communicating with other vehicle pairs or the central node.

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In this way, the data exchange overhead can be efficiently reduced.

- Utilizing the effective capacity theory, we derive the URLLC constraint using the large-scale CSI and formulate the large-scale resource allocation problem at the central node. To enhance the system performance at the occurrence of extreme events, a hybrid spectrum allocation strategy is proposed which enables spectrum sharing among different vehicle pairs. We decouple the formulated problem into several subproblems and a greedy-based searching algorithm is proposed to find the optimal solution.
- Based on the resource allocation result from the central node, a small-scale risk-aware constraint is modeled based on the instantaneous CSI and queue state of V2V pairs. Afterward, the small-scale resource allocation problem is formulated to minimize the power consumption while satisfying the URLLC requirements. Finally, the problem is transformed into a convex optimization problem and the optimal solution is obtained.

The rest of this paper is organized as follows. Section II introduces the system model and proposes the novel two time-scale resource allocation framework. Section III formulates the large-scale resource optimization problem and derives the optimal algorithm. In Section IV, the small-scale resource optimization problem is formulated and the corresponding optimal solution is devised. In Section V, simulation results are provided to demonstrate the effectiveness of the proposed algorithm. Finally, the article is concluded in Section VI.

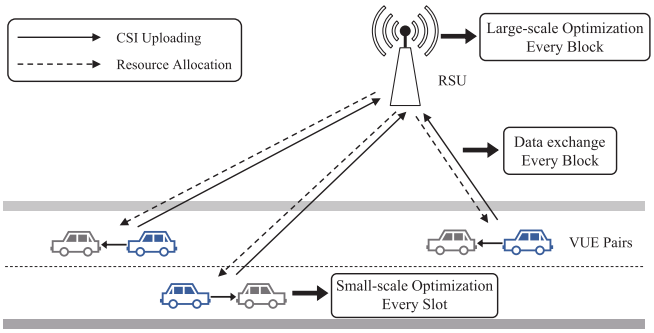
II. System Model and Resource Allocation Framework

In this section, we first present the system scenario, transmission model, and queuing model of the V2V network. After that, a resource allocation framework is proposed to guarantee the latency and reliability requirements of V2V links.

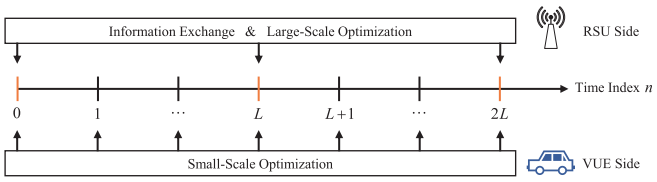
A. System Scenario

As shown in Fig. 1a, we consider a V2V network in the D2D-enabled cellular system, where K pairs of proximate vehicles, denoted as vehicle users (VUEs), are performing D2D communications with the assist of a single roadside unit (RSU). The slotted transmission model is considered where the slots are indexed by n, each with a duration of T. The total system bandwidth is B_t and is divided into M orthogonal RBs with bandwidth $B = B_t/M$. The sets of VUEs and RBs are denoted by $\mathcal{K} = \{1, 2, \ldots, K\}$ and $\mathcal{M} = \{1, 2, \ldots, M\}$, respectively. Assume that each VUE is equipped with a single antenna and an infinite-size first-in first-out queue which stores the data to be transmitted. In this work, we assume that the transmissions of the V2V pairs have already been synchronized. In addition, the time synchronization accuracy is accurate enough so that the transmissions are not impacted by the synchronization bias.[1]

[1]The accuracy achieved by GNSS-based time synchronization is better than 40 ns for more than 95% of time [36], which is much shorter than the cyclic prefix of OFDM symbols.



(a) A vehicular network with multiple VUE pairs assisted by a single RSU.



(b) Timeline of the two-timescale resource allocation framework.

Fig. 1. The vehicular network and two-timescale resource allocation framework.

B. Transmission Model

Let $h_k^m(n) = \rho_k g_k^m(n)$ represent the channel gain of the m-th RB which is occupied by VUE k during slot n, where ρ_k and $g_k^m(n)$ are large-scale and small-scale channel fading, respectively. To be specific, the large-scale channel fading is given by $\rho_k = G\psi_k d_k^{-\phi}$, where G is the pathloss constant, ψ_k denotes the log-normal shadowing random component at VUE k, d_k represents the distance between VUE pair k, and ϕ denotes the path loss exponent. Let L consecutive time slots form a time block. We then assume that large-scale channel fading ρ_k remains constant during a time block since the locations of the VUEs do not change too much during L time slots. As for the small-scale channel fading, Rayleigh fading is assumed for all VUEs. To be specific, $g_k^m(n)$ is independent and exponentially distributed (i.i.d) with unit mean for $\forall k \in \mathcal{K}$ and $\forall m \in \mathcal{M}$ over different slots. We note that this assumption is adopted for analytical tractability, while Rician fading model is more accurate in practical systems when there are strong line-of-sight (LOS) paths in V2V links. Nevertheless, the proposed framework and algorithm can be still applied to the Rician fading model.

In order to reduce the data exchange overhead, the RBs are periodically allocated by the RSU in the proposed resource allocation framework. During each block, the RB allocation remains unchanged since the VUEs cannot communicate with the RSU or other VUEs. If all RBs are allocated orthogonally, the VUEs can only transmit data on the allocated RBs and the latency requirement may not be guaranteed when the allocated RBs experience deep fading or the instantaneous data arrival rate is large. On the other hand, if some RBs are reserved as the shared set, the VUEs can occupy the shared RBs when the allocated orthogonal RBs cannot satisfy the latency requirement. When the number of available RBs is limited, it is hard to allocate a large number of RBs to each VUE. Nevertheless, it is possible to reserve a number of RBs as the

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shared set, which can be randomly occupied by the VUEs to enhance the URLLC performance. Therefore, in this paper, we consider a hybrid RB occupation mode where M RBs are divided into an orthogonal set and a shared set. The RBs in the orthogonal set are orthogonally allocated to the VUEs and N_k denotes the number of RBs allocated to VUE k. In contrast, the RBs in the shared set are shared by all VUEs. The shared set is denoted by $\mathcal{U}$ which is composed by U RBs. At the occurrence of extreme events such as deep channel fading, when the allocated RBs in the orthogonal set cannot guarantee the URLLC requirements, the VUEs will choose the RBs from the shared set to transmit data. If a certain RB is chosen by more than one VUE simultaneously, the transmissions on this RB will collide and the data cannot be correctly decoded by the receivers.

C. Resource Allocation Framework

Due to the strict latency requirement of V2V transmissions, the procedure which increases the transmission delay needs to be avoid. In particular, the resource allocation algorithms which require frequent CSI exchange are not practical for low-latency V2V communications, since the latency caused by CSI exchange overhead is non-negligible. To reduce CSI exchange overhead, the large-scale channel information of all V2V links is collected at the central node to model the time-average URLLC performance, leveraging the theory of effective capacity [33]. However, the work in [33] assumes that the data arrival rate of the VUEs is constant, which cannot be applied to Poisson arrival model due to the uncertain data arrival rate. To deal with this mismatch and guarantee the URLLC performance, the accurate information of the VUEs, such as the real-time CSI and the instantaneous queue state, should be utilized to formulate a small-scale risk-aware resource allocation problem at each VUE.

In this paper, we propose a two time-scale resource allocation framework to reduce the information exchange overhead while guaranteeing system performance. As shown in Fig. 1b, the framework employs a large-scale resource optimization and a small-scale resource optimization. The large-scale resource optimization is conducted at the RSU every time block which only requires the large-scale channel information ρ_k and the average data arrival rates of all VUEs. Since the RSU-VUE communication is only conducted every block and only the average information is needed, the data exchange overhead in this phase is negligible. After that, the RB and power allocation result is fed back to the VUEs. It is noted that the power allocation is a preliminary result which will be further optimized at the VUEs. At the VUE side, it is assumed that the instantaneous channel state is estimated by the receiver and fed back to the transmitter every time slot. In this way, each VUE pair can obtain the small-scale channel information $h_k^m(n)$. According to the large-scale resource allocation result, the instantaneous CSI, and the queue state, each VUE pair performs the small-scale resource allocation every time-slot without communicating with other VUE pairs. The aim of small-scale resource allocation is to determine the instantaneous RB choice and power allocation. In the following sections, we will discuss the details of the large-scale and small-scale resource allocation problems, respectively.

III. Large-Scale Resource Allocation

In this section, we leverage the effect capacity theory to model the latency constraint which only requires the large-scale channel information of the VUEs. After that, the impact of RB sharing is analyzed and the large-scale resource allocation problem is formulated at the RSU to minimize the transmit power while guaranteeing the URLLC requirements. Finally, the optimal solution to the large-scale resource allocation problem is derived.

A. Large-Scale URLLC Model

In this work, the latency violation probability (LVP) [9], [33] is employed as the URLLC metric of the vehicular communication, which characterizes the tail behavior of random latency. Statistically, LVP is the proportion of expired data among all data. To be specific, the expired data includes both the delayed data that has been stored in the queue for more than $D_{\max}$ slots and the collided data caused by RB collisions. Due to the randomness of channel fading and data arrival process, the expression of LVP with respect to system parameters is hard to be derived. Therefore, we introduce the effective capacity theory which characterizes the quality of wireless links by the QoS exponent θ.

It is assumed that the VUEs can obtain the data arrival rate λ_k and the large-scale CSI ρ_k at the beginning of each block, which are then transmitted to the RSU [33]. Denote P_k^m as the power allocated to the m-th RB occupied by VUE k. Then the channel capacity of VUE k can be expressed as

$$C_k = \sum_{m=1}^{N_k} B\log_2\left(1+\frac{P_k^m \rho_k g_k^m}{N_0 B}\right), \tag{1}$$

where g_k^m is the random small-scale Rayleigh fading of the m-th RB of VUE k. The shared RBs are not included in the expression of the channel capacity since they are only occupied during the extreme events such as deep channel fading and queue overflow.

In order to model the LVP performance of the VUEs, we first calculate the effective capacity of V2V link k as [37]

$$EC_k(\theta_k) = -\frac{1}{\theta_k T}\ln \mathbb{E}\left\{e^{-\theta_k T C_k}\right\}, \tag{2}$$

where $\mathbb{E}$ indicates the expectation operator and $\theta_k \geqslant 0$ is the latency exponent of VUE k. In particular, $EC_k(\theta_k)$ accounts for the maximum data arrival rate that VUE k can support subject to latency exponent θ_k and is a monotonically decreasing function of θ_k [38]. When $\theta_k = 0$, there is no guarantee on LVP and $EC_k(\theta_k = 0) = \mathbb{E}\{C_k\}$ is the ergodic capacity. As θ_k becomes larger, the latency constraint is more strictly guaranteed at the expense of reduced effective capacity. When $\theta_k = \infty$, the system cannot tolerate any delay and $EC_k(\theta_k = \infty) = 0$. When the effective capacity is equal to the data arrival rate, as

$$EC_k(\theta_k) = \lambda_k, \tag{3}$$

the LVP of VUE k can be expressed as

$$P_{\mathrm{D},k} \approx \mu(\lambda_k)\, e^{-\theta_k \lambda_k T D_{\max}} \approx \frac{\lambda_k}{\mathbb{E}\{C_k\}} e^{-\theta_k \lambda_k T D_{\max}}, \tag{4}$$

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where $\mu(\lambda_k) \approx \frac{\lambda_k}{\mathbb{E}\{C_k\}} \leqslant 1$ is the probability that queue k is not empty [37]. The inequality always holds since $\lambda_k = EC_k(\theta_k) \leqslant EC_k(0) = \mathbb{E}\{C_k\}$. In fact, θ_k is an intermediate variable which can be eliminated by solving equations (2)-(4). As a result, $P_{\mathrm{D},k}$ is determined by the number of RBs, N_k, and the transmit power P_k^m allocated to VUE k. For simplicity, it is assumed that all the VUEs have the same LVP requirement P_{D} and the RSU can adjust the RB and power allocation to ensure that $P_{\mathrm{D},k} = P_{\mathrm{D}}, \forall k \in \mathcal{K}$. The detailed derivation of the effective capacity and LVP expressions can be found in [33], [37], [38]. It is noted that the infinite-size memory is a basic assumption of the effective capacity theory, which ignores packet drop due to buffer overflow to obtain the closed-form LVP expression (4). However, the buffer size is usually limited in practice. When considering packet drop due to buffer overflow, the actual LVP would be smaller than $P_{\mathrm{D},k}$, because the dropped packets do not need to be transmitted and the dynamic queue length becomes shorter. This is true when the buffer size is larger than $\lambda T D_{max}$, which guarantees that the dropped packets have already violated the latency constraint. Therefore, (4) can be regarded as an upper bound of the actual LVP and the gap is minor when the memory size is sufficiently large.

During the above analysis, the LVP expression (4) only considers the orthogonal RBs occupied by the VUEs. Now we consider the impact of the shared RBs. To do so, we assume that a VUE will choose a RB from U shared RBs to transmit data when the transmission on the orthogonal RBs cannot satisfy the latency requirement with the probability P_{D}. Let P_{C} denote the RB collision probability, i.e., the probability that more than one VUE choose the same RB from $\mathcal{U}$. Based on the probability theory, P_{C} can be expressed as

$$P_{\mathrm{C}} = \sum_{i=1}^{K-1}\left[1-\left(\frac{U-1}{U}\right)^i\right] C(K-1,i)\left(P_{\mathrm{D}}\right)^i\left(1-P_{\mathrm{D}}\right)^{K-1-i} \approx \frac{K-1}{U}P_{\mathrm{D}}. \tag{5}$$

where $C(K-1,i)$ is the binomial coefficient. In other words, there are K balls and U blankets, each ball is picked up with the probability P_{D} and thrown into a random blanket. Then P_{C} is the probability that more than one ball falls into the same blanket. Since the LVP requirement P_{D} is close to zero according to the URLLC requirements, the above approximation holds by omitting the items with orders higher than one. The latency constraint will be finally violated when a VUE chooses a RB from the shared set and the RB encounters a collision. Therefore, by taking into account the RB collision, the final LVP of the vehicular system is expressed as

$$P_{\mathrm{LVP}} = P_{\mathrm{D}}P_{\mathrm{C}} \approx \frac{K-1}{U}P_{\mathrm{D}}^2. \tag{6}$$

B. Problem Formulation

Since the maximum transmit power of the VUEs is limited, we aim at minimizing the maximum power consumption of all VUEs while guaranteeing the URLLC requirements, where the min-max objective function ensures user fairness. Holding the extra power budget, the VUEs can effectively reduce the transmission LVP at the occurrence of extreme events. Therefore, the large-scale resource allocation problem can be formulated as

$$\mathcal{P}1: \quad \min_{\left\{P_k^m, N_k, U\right\}} \ \max_k \left(\sum_{m=1}^{N_k} P_k^m\right), \tag{7a}$$

$$\text{s.t. } \frac{K-1}{U}P_{\mathrm{D}}^2 \leqslant \gamma, \tag{7b}$$

$$\sum_{m=1}^{N_k} P_k^m \leqslant P_{\max}, \quad \forall k \in \mathcal{K}, \tag{7c}$$

$$P_k^m \geqslant 0, m = 1, \ldots, N_k, \quad \forall k \in \mathcal{K}, \tag{7d}$$

$$\sum_{k=1}^{K} N_k + U = M, \tag{7e}$$

$$N_k \in Z^+, \quad U \in Z^+, \tag{7f}$$

where (7b) restricts the LVP of the VUEs according to the effective capacity theory and the collision model with $\gamma > 0$ being the LVP threshold of the vehicle system which is very small, (7c), (7d), and (7f) are the feasible constraints, $P_{\max}$ is the power budget of the VUEs,[2] and (7e) is the RB number constraint. The optimization variables are the power allocation P_k^m and the RB allocation $\{N_k, U\}$.

C. Optimal Solution

$\mathcal{P}1$ is a mixed-integer non-linear programming problem which is difficult to solve directly. Therefore, we will first decompose the problem into K subproblems by fixing the RB allocation $\{U, N_k\}$ and derive the optimal power allocation P_k^m. Then, we develop an algorithm to find the optimal RB allocation through separately optimizing the numbers of orthogonal and shared RBs. We first introduce the following theorem.

Theorem 1: When U and N_k are given, the optimal solution to $\mathcal{P}1$ can be obtained by solving K subproblems $\mathcal{P}2_k, k \in \mathcal{K}$.

$$\mathcal{P}2_k: \quad \min_{P_k^m} \sum_{m=1}^{N_k} P_k^m, \tag{8a}$$

$$\text{s.t. } \frac{K-1}{U}P_{\mathrm{D}}^2 \leqslant \gamma, \tag{8b}$$

$$\sum_{m=1}^{N_k} P_k^m \leqslant P_{\max}, \tag{8c}$$

$$P_k^m \geqslant 0, \quad m = 1, \ldots, N_k. \tag{8d}$$

Proof: When $\{U, N_k\}$ are given, the constraints of different subproblems are independent of each other. It is also easy to prove that the objective function of $\mathcal{P}1$ is minimized when the sum power of each VUE is minimized. ∎

After decomposing problem $\mathcal{P}1$ into K subproblems, we will derive the optimal solution to each subproblem according to the following theorem.

Theorem 2: The optimal solution to subproblem $\mathcal{P}2_k$ is allocating equal power to each RB, i.e., $P_k^m = \frac{\sum_{m=1}^{N_k} P_k^m}{N_k}, m = 1, \ldots, N_k$.

[2]The power budget is assumed to be the same for all VUEs, which is the common case in practical vehicular systems. Nevertheless, our model and analysis can be directly generalized to the scenario where different vehicle pairs have different power budgets.

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Algorithm 1 The Optimal Solution to $\mathcal{P}1$

1: **for** each $U \in [1, M-K]$ **do**
2: Set $N_k = 1, \forall k \in \mathcal{K}$;
3: **while** $\sum_{k=1}^{K} N_k < M - U$ **do**
4: Solve subproblems $\mathcal{P}2_k, \forall k \in \mathcal{K}$ and get the optimal solution P_k^m;
5: Find the VUE k_0 that has the maximum sum power $\sum_{m=1}^{N_{k_0}} P_{k_0}^m$ among all VUEs;
6: Perform $N_{k_0} \leftarrow N_{k_0} + 1$;
7: **end while**
8: **end for**
9: Output the optimal $\{U^*, N_k^*, P_k^{m*}\}$ that minimize the objective function of $\mathcal{P}1$.

Proof: Please see Appendix A. ∎

Therefore, to solve subproblem $\mathcal{P}2_k$, we only need to optimize the sum power $P_{\text{sum}} = \sum_{m=1}^{N_k} P_k^m$ and equally allocate the power to each RB as $P_k^m = \frac{P_{\text{sum}}}{N_k}$. It is easy to prove that P_{D} decreases with P_{sum} and the optimal solution satisfies that $P_{\text{D}} = \sqrt{\frac{U\gamma}{K-1}}$ which means that P_{sum} can be calculated through binary search. Therefore, we can easily get the optimal solution to $\mathcal{P}2_k$ by performing bisection search on P_{sum}.

In conclusion, the optimal solution to problem $\mathcal{P}1$ can be obtained through Algorithm 1. In the algorithm, we perform exhaustive search on the number of shared RBs U over $U \in [1, M-K]$. As for the allocation of orthogonal RBs N_k, a greedy algorithm can be applied to find the optimal N_k by searching $\sum_{k=1}^{K} N_k$ over $\sum_{k=1}^{K} N_k < M - U$, whose optimality is proved in Appendix B. At each step of the search, problem $\mathcal{P}2_k$ is solved once and the computational complexity is $O(\log(d))$, where $d = P_{\max}/d_{\min}$ and $d_{\min}$ is the granularity of binary search. Therefore, the total computational complexity is $O(M^2 \log(d))$. The solution to the large-scale resource allocation problem which is conducted at the RSU will be transmitted to all VUEs as the parameters of the small-scale resource allocation problem, which will be discussed in the following section.

IV. Small-Scale Resource Allocation

Although the communication overhead can be efficiently reduced by performing the large-scale resource allocation once in a long period, instantaneous information including accurate CSI and queue state remains under-utilized, which may lead to performance degradation compared with some centralized algorithms utilizing more information. Moreover, due to the restrict of effective capacity theory, the data arrival rate is assumed to be constant during the large-scale resource allocation. As a result, the queue dynamic brought by the uncertain data arrival rate at the VUEs cannot be captured and the actual LVP performance at the VUEs will degrade. To make up for the above deflects, the small-scale resource allocation at the VUEs is proposed in this section to improve system performance by utilizing instantaneous channel and queuing information. Firstly, we design a small-scale URLLC metric based on the instantaneous channel information and queue state. After that, a transmission utility expression is proposed to measure the effectiveness of data transmission. During each time slot, the small-scale resource allocation problem is formulated at each VUE to reach the tradeoff between power consumption and transmission utility while guaranteeing the URLLC requirements. Finally, the optimal solution to the small-scale resource allocation problem is derived.

A. Queuing Model

As shown in Fig. 2, the transmitter of each VUE has an infinite-size first-in first-out queue which stores the data to be transmitted.[3] Different from the existing works which only consider the amount of data in the queue, we focus on different queuing delays of different data in the queue. To be more specific, $Q_k^i(n)$ denotes the amount of data which has been stored for i slots in queue k at slot n before transmission. Besides, let $Q_k(n) = \sum_{i=0}^{D_{\max}} Q_k^i(n)$ denote the total amount of data in queue k. It is noted that the data that has been stored for more than $D_{\max}$ slots violates the latency constraint and would be dropped from the queue. Let $A_k(n)$ denote the amount of data arriving at VUE k in slot n, where Poisson arrival process with average arrival rate λ_k is assumed for all VUEs. Let $S_k(n) = TC_k(n)$ denote the amount of transmitted data at VUE k during slot n. The channel capacity of VUE k during slot n is given by

$$C_k(n) = \sum_{m=1}^{N_k^* + U_k(n)} B\log_2\left(1 + \frac{p_k^m(n) h_k^m(n)}{N_0 B}\right), \tag{9}$$

where N_k^* is the RB allocation result of the large-scale optimization problem, $U_k(n)$ represents the number of shared RBs occupied by VUE k during slot n which follows the constraint $U_k(n) \leqslant U^*$, and $p_k^m(n)$ denotes the power allocated to the m-th RB at VUE k during slot n. It is noted that $p_k^m(n)$ is the slot-based power allocation which is different from the block-based power allocation P_k^{m*}. Since the RSU can only leverage the large-scale information, P_k^{m*} is actually a preliminary power allocation result which needs to be further optimized at the VUEs. The final power allocation $p_k^m(n)$ is optimized by the small-scale optimization, taking into account both the instantaneous channel information and the queue state.

The queue dynamic at VUE k can then be expressed as

$$Q_k^i(n+1) = \begin{cases} 0, & w+2 \leqslant i \leqslant D_{\max}, \\ \sum_{i=w}^{D_{\max}} Q_k^i(n) - TC_k(n), & i = w+1, \\ Q_k^{i-1}(n), & 1 \leqslant i \leqslant w, \\ A_k(n+1), & i = 0, \end{cases} \tag{10}$$

[3]The infinite-size memory is assumed in the small-scale optimization to follow the assumption of the effective capacity theory. Nevertheless, the small-scale optimization can also be applied to the case when each VUE is equipped with a finite-size buffer.

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Fig. 2. The queuing model of VUE k.

where w represents the maximum queuing latency of the data in the queue after transmission and is given by

$$w = \underset{l}{\text{argmax}} \left(\sum_{i=l}^{D_{\max}} Q_k^i(n) - TC_k(n) \right),$$
$$\text{s.t.} \sum_{i=l}^{D_{\max}} Q_k^i(n) - TC_k(n) > 0,$$
$$l = 0, \ldots, D_{\max}. \tag{11}$$

In other words, the initial queuing latency of the newly arrived data is zero and the queuing latency of the remaining data in the queue after transmission will increase by one at the next slot.

B. Small-Scale URLLC Model

Let $\hat{Q}_k^i(n), i = 0, \ldots, D_{\max}$, denote the amount of data which has been stored for i slots in queue k at slot n after transmission. To be specific,

$$\hat{Q}_k^i(n) = \begin{cases} 0, & w+1 \leqslant i \leqslant D_{\max}, \\ \sum_{i=w}^{D_{\max}} Q_k^i(n) - TC_k(n), & i = w, \\ Q_k^i(n), & 0 \leqslant i \leqslant w-1. \end{cases} \tag{12}$$

Then we define the small-scale delay violent probability of VUE k at slot n as

$$P_{\text{delay}}(n) = \frac{\sum_{i=0}^{D_{\max}} \hat{Q}_k^i(n) \Pr\left\{ \sum_{l=1}^{D_{\max}-i} TC_k(n+l) \leqslant \sum_{l=i}^{D_{\max}} \hat{Q}_k^l(n) \right\}}{\sum_{i=0}^{D_{\max}} Q_k^i(n)}, \tag{13}$$

where $\Pr\left\{ \sum_{l=1}^{D_{\max}-i} TC_k(n+l) \leqslant \sum_{l=i}^{D_{\max}} \hat{Q}_k^l(n) \right\}$ is the probability that the data $\hat{Q}_k^i(n)$ cannot be completely transmitted in the following $D_{\max} - i$ slots. In fact, $P_{\text{delay}}(n)$ is the average LVP of the data in the queue and can be utilized as the URLLC metric of the VUEs. However, the expression of $P_{\text{delay}}(n)$ is implicit which cannot be applied to the optimization problem directly. Therefore, we need to derive the explicit expression of $P_{\text{delay}}(n)$ with respect to the resource allocation result at the current slot.

To simplify the expression of $P_{\text{delay}}(n)$, we first denote the channel capacity of the m-th RB at VUE k as

$$C_k^m = B \log_2 \left(1 + \frac{P_k \rho_k g_k^m}{N_0 B} \right), \tag{14}$$

where $P_k = P_k^{m*}, m = 1, \ldots, N_k^*$, is the power allocation result of the RSU. Then the cumulative distribution function (CDF) of C_k^m can be approximated by

$$\begin{aligned} \Pr\{C_k^m \leqslant y\} &= \Pr\left(B \log_2 \left(1 + \frac{P_k \rho_k g_k^m}{N_0 B} \right) \leqslant y \right) \\ &= \Pr\left(g_k^m < \frac{N_0 B}{\rho_k P_k} \left(2^{\frac{y}{B}} - 1 \right) \right) \\ &= 1 - e^{-\frac{N_0 B}{\rho_k P_k} \left(2^{\frac{y}{B}} - 1 \right)} \\ &\approx 1 - e^{-\frac{N_0 \ln 2}{\rho_k P_k} y}. \end{aligned} \tag{15}$$

The above approximation holds since we only focus on the extreme events when the channel state is poor and the channel capacity y is very small. Therefore, C_k^m approximately follows the exponential distribution with parameter $\frac{N_0 \ln 2}{\rho_k P_k}$ and $\sum_{l=1}^{D_{\max}-i} C_k(n+l)$ approximately follows the gamma distribution, i.e., $Gamma\left(N_k^*(D_{\max} - i), \frac{N_0 \ln 2}{\rho_k P_k} \right)$. Denote $Y_k^i = \sum_{l=1}^{D_{\max}-i} C_k(n+l)$ as the gamma distributed random variable. Then (13) can be approximated as

$$P_{\text{delay}}(n) = \frac{\sum_{i=0}^{D_{\max}} \hat{Q}_k^i(n) \Pr\left\{ TY_k^i \leqslant \sum_{l=i}^{D_{\max}} \hat{Q}_k^l(n) \right\}}{\sum_{i=0}^{D_{\max}} Q_k^i(n)}. \tag{16}$$

It is easy to prove that $\hat{Q}_k^i(n)$ is a monotonically non-increasing function of $C_k(n)$ and the value of $\Pr\left\{ \sum_{l=1}^{D_{\max}-i} TY_k^i \leqslant \sum_{l=i}^{D_{\max}} \hat{Q}_k^l(n) \right\}$ is a monotonically non-decreasing function of $\sum_{l=i}^{D_{\max}} \hat{Q}_k^l(n)$. Thus $P_{\text{delay}}(n)$ is a monotonically non-increasing function of $C_k(n)$. Therefore, the URLLC constraint $P_{\text{delay}}(n) \leqslant \gamma$ can be identically converted into the transmission rate constraint $C_k(n) \geqslant C_k^{\min}(n)$, where $C_k^{\min}(n)$ can be obtained by letting $P_{\text{delay}}(n) = \gamma$.

C. Problem Formulation

Different from the large-scale resource allocation problem which aims at minimizing the transmit power, the small-scale resource allocation problem is formulated at each VUE during each slot to reach the tradeoff between power consumption and transmission utility while guaranteeing the URLLC requirements. Adding the transmission utility into the objective function implies that the transmission rate increases with channel gain. In other words, the VUE will transmit more data to shorten the queue length when the channel condition is good and transmit less data to reduce the power consumption when the channel condition is poor. In this way, the long-term power efficiency can be improved.

The transmission utility represents the reduction of total queuing time of the data in the queue brought by the data transmission, which can be defined as

$$J(C_k(n)) = \sum_{i=0}^{D_{\max}} \left(iQ_k^i(n) - i\hat{Q}_k^i(n) \right). \tag{17}$$

Therefore, the small-scale resource allocation problem can be formulated as

$$\mathcal{P}3: \quad \min_{\{p_k^m(n), U_k(n)\}} \sum_{m=1}^{N_k^* + U_k(n)} p_k^m(n) - \alpha J(C_k(n)) + \beta U_k(n), \tag{18a}$$
$$\text{s.t. } C_k(n) \geqslant C_k^{\min}(n), \tag{18b}$$
$$\sum_{m=1}^{N_k^*} p_k^m(n) \leqslant P_{\max}, \tag{18c}$$
$$U_k(n) \leqslant U^*, U_k(n) \in Z^+ \tag{18d}$$

where α is the tradeoff parameter, β is a large positive constant to minimize the number of occupied shared RBs to reduce

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the collision probability, (18b) is the URLLC constraint, (18c) is the power budget constraint, and (18d) is the feasible constraint. It is noted that, apart from the large-scale resource allocation result from RSU, only the local CSI and queue state at VUE k are required in problem $\mathcal{P}3$, which implies that the communications among VUE pairs are not needed.

D. Optimal Solution

From the objective function (18a), the large value of β implies that the optimal solution can be obtained by the following strategy. Firstly, set $U_k(n) = 0$ and find the optimal solution. While there is no feasible solution, we choose a shared RB with the largest channel gain from the shared RBs and set $U_k(n)$ to be $U_k(n) + 1$ until there exists a feasible solution or $U_k(n) = U^*$.[4] Without loss of generality, we let $M_k(n) = N_k^* + U_k(n)$ and omit the third term of the objective function, i.e., $\beta U_k(n)$, in the following derivations, as $U_k(n)$ can be regarded as a constant when solving the problem according to the above strategy.

Theorem 3: The optimal solution to $\mathcal{P}3$ satisfies that $p_k^m(n) = \max\{\eta - \sigma_k^m(n), 0\}$, where $\sigma_k^m(n) = \frac{N_0 B}{h_k^m(n)}$ is the equivalent channel noise and η is a constant which can be calculated by substituting $p_k^m(n)$ into (9) when $C_k(n)$ is given. In other words, $p_k^m(n)$ is the water pouring power allocation which is an implicit function of the channel capacity $C_k(n)$.

Proof: Please see Appendix C. ∎

Therefore, we can add a constraint that $C_k(n) = G_k\left(\sum_{m=1}^{M_k(n)} p_k^m(n)\right)$ without changing the optimal solution to $\mathcal{P}3$, where $G_k(x)$ represents channel capacity achieved by water pouring power allocation with the sum power $\sum_{m=1}^{M_k(n)} p_k^m(n) = x$. Since $G_k(x)$ is a monotonically increasing function of x which implies that the inverse function exists, problem $\mathcal{P}3$ can be further transformed into

$$\mathcal{P}4: \quad \min_{x} \; x - \alpha J\left(G_k(x)\right), \tag{19a}$$

$$\text{s.t. } x \geqslant G_k^{-1}\left(C_k^{\min}(n)\right), \tag{19b}$$

$$x \leqslant P_{\max}, \tag{19c}$$

which is a one-dimensional optimization problem with a single variable x, and can be solved by exhaustive searching.

Theorem 4: $H(x) = x - \alpha J\left(G_k(x)\right)$ is a convex function.

Proof: Please see Appendix D. ∎

To sum up, the original problem $\mathcal{P}3$ can be transformed into the convex problem $\mathcal{P}4$ with a single optimization variable x. The optimal sum power x can be obtained by performing binary search over $x \in [0, P_{\max}]$. After that, according to Theorem 3, the optimal power allocation $p_k^m(n)$ of problem $\mathcal{P}3$ can be obtained through the water pouring algorithm with the sum power x. Therefore, the overall computational complexity of the small-scale optimization is $O(\log(d))$.

V. Simulation Results

In this section, simulations are conducted to validate the proposed resource allocation algorithms for V2V networks.

[4]Since the small-scale channel gain is independent and exponentially distributed with unit mean for all VUE pairs, the order of shared RBs in terms of the channel gain is independent for all VUE pairs. Therefore, the proposed RB selection strategy does not increase the collision probability of data transmission.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Number of VUE pairs, K	20
Number of RBs, M	50
RB bandwidth, B	180 KHz
Slot length, T	1 ms
Block length, L	1000
Maximum transmit power, $P_{\max}$	17 dBm
Power spectral density of noise, N_0	-174 dBm/Hz
Number of lanes	4
Lane width	4 m
Absolute vehicle speed	70 km/h
Vehicle density	Average inter-vehicle distance is 2.5 sec × absolute vehicle speed.
Average data arrival rate, λ	0.5 Mbps
Latency constraint, $D_{\max}$	5
Reliability constraint, γ	10^{-5}

We model a multi-lane freeway which passes an RSU as illustrated in Fig. 1a. The vehicles are deployed on the lanes according to spatial Poisson process and the VUEs pairs are randomly generated from the neighboring vehicles. The average data arrival rate λ is assumed to be the same for all VUEs. The simulation setup follows the details in [31], [39] and the major parameters are listed in Table I. The path-loss model is adopted as $128.1 + 37.6\log(d \text{ [km]})$ and log-normal shadowing distribution with standard deviation of 8 dB and Rayleigh fast fading are also assumed. It is noted that all the parameters are set as Table I by default unless otherwise specified and the simulation results are obtained by averaging over 10,000 channel realizations.

To demonstrate the performance of the proposed method, we also test the methods in [16] and [20] as baselines, respectively. In [16], leveraging EVT, the latency violation probability is modeled by the entropic risk which is related to the maximal queue length. Firstly, the RBs are allocated to VUEs by the RSU according to the geographic proximity information. Then the VUEs approximately characterize the distribution of queue length which excesses a certain value using the generalized Pareto distribution (GPD). Finally, the power minimization problem is solved by the Lyapunov optimization by introducing two virtual queues which impose the constraints on the mean and second moment of the conditional queue length. In [20], the AoI is utilized to model the latency performance of the vehicular system. With the global instantaneous CSI and queue state, the RSU dynamically allocates the RBs and the transmit power of all VUEs every slot. Similarly, the time-averaged constraints are converted to several virtual queues and the power minimization problem is solved by the Lyapunov optimization. It is noted that, the AoI-based method considers the ideal case when the RSU is aware of the instantaneous information of all VUEs, which is expected to provide better latency performance and higher power efficiency at the cost of heavy data exchange overhead.

Fig. 3 shows the average LVP of the proposed method and the baseline methods with varying number of VUEs, where the latency constrains $\tau = 3$ ms and $\tau = 5$ ms are considered. It can be observed that the LVP increases with the number since more VUEs would aggravate the spectrum usage and reduce the frequency diversity. Moreover, the LVP becomes

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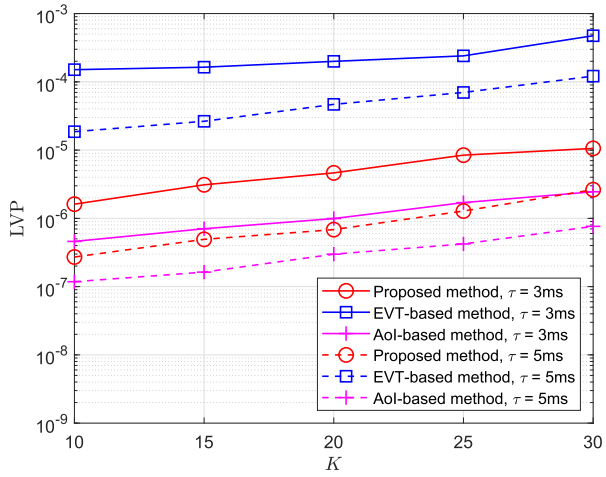


Fig. 3. Average LVP of VUEs with varying number of VUEs.

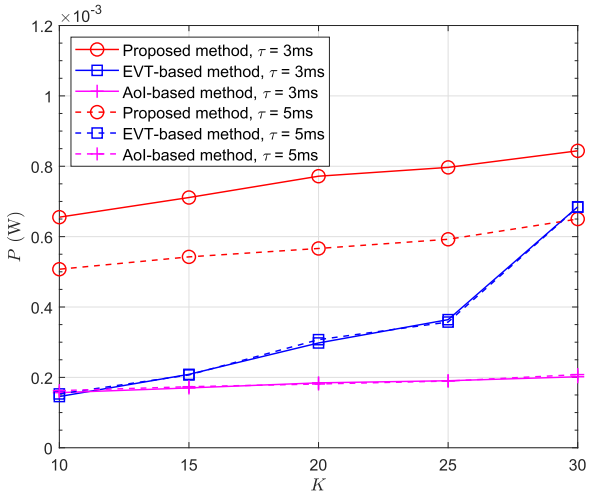


Fig. 4. Average transmit power of VUEs with varying number of VUEs.

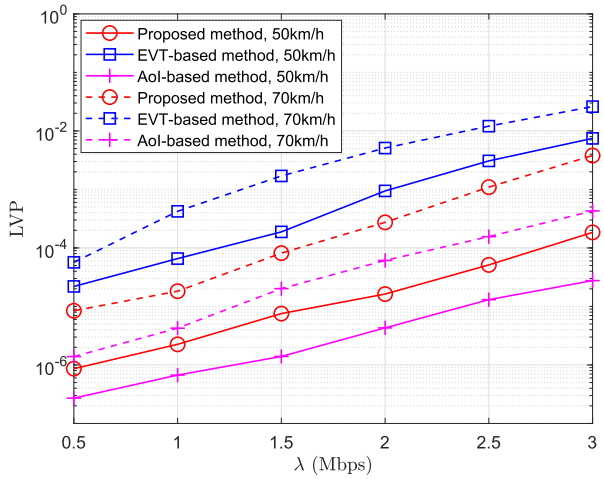


Fig. 5. Average LVP of VUEs with varying data arrival rates.

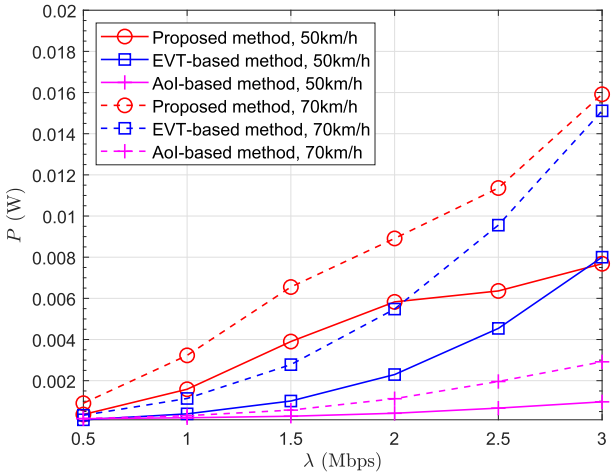


Fig. 6. Average transmit power of VUEs with varying data arrival rates.

smaller with a larger τ since the latency requirement of VUEs becomes less stringent. It is also observed that, benefiting from the proposed resource allocation framework, our method provides a much lower LVP than the EVT-based method with different VUE numbers and different latency constraints. Meanwhile, the AoI-based method achieves the lowest LVP value since the RSU utilizes the global instantaneous CSI and queue state to guarantee the latency performance at each slot.

The impact of VUE number and latency constraint on the average transmit power of VUEs is depicted in Fig. 4, where the latency constraints of 3 ms and 5 ms are compared. From the figure, the average transmit power increases with the number of VUEs since more VUEs would aggravate the spectrum usage and reduce the energy efficiency as mentioned previously. It is noted that the average transmit power of the baseline methods is almost the same for different latency constraints since the baseline methods do not consider the latency constraint τ as the parameter and only model the latency violation risk according to the queue length. In comparison, the latency constraint τ is considered in the proposed method. It can be observed that the average transmit power becomes larger when τ is smaller because smaller τ indicates stricter latency constraint and requires larger transmit power to reduce the queue length. Compared with the EVT-based method, the proposed method has lower power efficiency since it reserves part of the spectrum resource as the shared RBs to achieve a smaller LVP. On the other hand, the AoI-based method achieves the highest power efficiency because the RSU can dynamically allocate the RBs and power according to the instantaneous CSI and queue state of all VUEs.

By varying the average data arrival rate, Fig. 5 depicts the LVP of the proposed method and the baseline methods with different vehicle speeds. As the average data arrival rate λ increases, the queue becomes more congested, which is illustrated in the figure as the LVP of both methods increases with λ. Meanwhile, the channel condition gets worse as the vehicle speed grows from 50 km/h to 70 km/h, leading to an increased LVP for both methods. It is also observed that, with different data arrival rates and vehicle speeds, the proposed method provides a much lower LVP than the EVT-based method, while the AoI-based method achieves the lowest LVP among all methods.

Fig. 6 demonstrates the average transmit power of VUEs for different average data arrival rates λ and different vehicle speeds. As λ increases, the average transmit power increases to support a larger transmission rate. We can also observe that the average transmit power increases with the vehicle speed. The reason is that a higher vehicle speed would result

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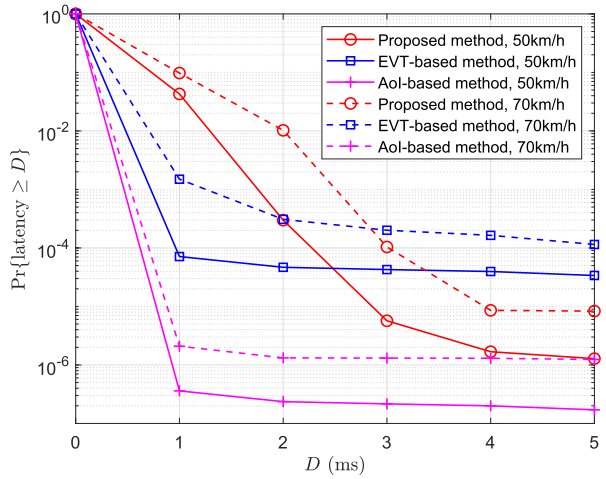


Fig. 7. The CCDFs of the data latency with different vehicle speeds.

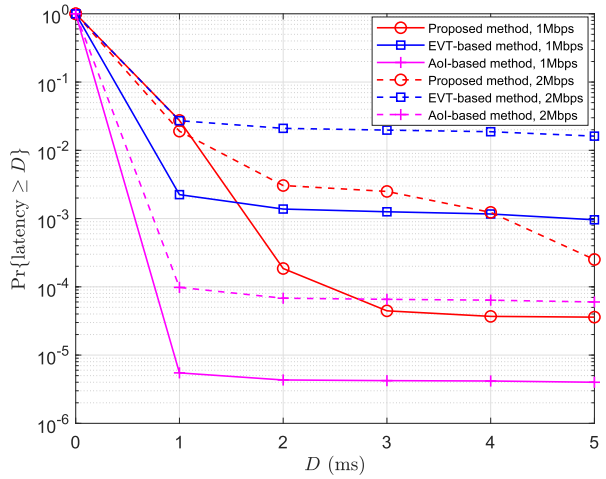


Fig. 8. The CCDFs of the data latency with different data arrival rates.

in a worse channel condition and finally reduces the power efficiency. It is noted that the power consumption of the proposed method is larger than the EVT-based method since the proposed method reserves part of the spectrum resource as the shared RBs to achieve a smaller LVP. However, the curves of the proposed method get close to those of the EVT-based method as λ increases. The reason is that more RBs are allocated for orthogonal transmission when λ is large, which increases the power efficiency of the proposed method.

The complementary cumulative distribution functions (CCDFs) of the data latency with different vehicle speeds are plotted in Fig. 7. Here the latency constrains is set to 5 ms, which corresponds to $D = 5$ in the figure. From the figure, a larger vehicle speed implies a worse channel condition and aggravates the congestion of queue, which increases the probability that the transmitted data has a relatively large latency. It is also observed that, the curves of the EVT-based method first drop rapidly and then gradually become flat. This is different from the curves of the proposed method which has a much smaller tail. According to the transmission utility (17) of the small-scale resource allocation, the proposed method tends to tolerate the data with small latency but is strict to the data with large latency. As a result, the proportion of data with small D of the proposed method is higher than that of the EVT-based method, which does not influence the LVP performance of the system. The above analysis also applies to Fig. 8, which shows the CCDFs of the data latency with different data arrival rates. Holding a smaller tail of latency distribution, the proposed method can better ensure the latency requirement than the EVT-based method. Again, the CCDF curve of the AoI-based method is the lowest among all methods, since the queue length is shortened by dynamically allocating the RBs and power according to the instantaneous CSI and queue state of all VUEs.

VI. Conclusion

This paper develops a novel hybrid resource allocation framework for the vehicular network to satisfy the URLLC requirements of V2V communications. To avoid the heavy communication overhead of exchanging instantaneous CSI, the large-scale resource allocation is performed at the RSU periodically, which determines the initial RB and power allocation leveraging the large-scale CSI collected from all VUEs. Meanwhile, the spectrum sharing scheme is considered to enhance the robustness of V2V transmission and improve the URLLC performance. Based on the large-scale resource allocation result received from the RSU, each VUE performs the small-scale resource allocation to determine the occupation of shared RBs and the instantaneous transmit power on each RB, utilizing the instantaneous CSI and queue state. With no requirement on the information of other VUEs, the small-scale resource allocation is conducted locally and the data exchange among VUEs is avoided. The optimal solutions to the formulated large-scale and small-scale resource allocation problems are derived, respectively. Finally, the simulation results show that the proposed resource allocation framework can satisfy the latency and reliability requirements of V2V networks.

Appendix A
Proof of Theorem 2

In order to prove Theorem 2, we first prove that both $f(x) = \mathbb{E}\left\{\log_2\left(1+\frac{h_k^m x}{N_0 B}\right)\right\}$ and $g(x) = -\frac{1}{\theta_k T}\ln \mathbb{E}\left\{e^{-\theta_k T B \log_2\left(1+\frac{h_k^m x}{N_0 B}\right)}\right\}$ are concave.

We rewrite $f(x)$ as

$$f(x) = \mathbb{E}\left\{\log_2\left(1+\frac{h_k^m x}{N_0 B}\right)\right\} = \int_0^\infty \log_2\left(1+\frac{\rho_k x \tau}{N_0 B}\right) e^{-\tau} d\tau. \quad (20)$$

Since $\log_2\left(1+\frac{\rho_k x \tau}{N_0 B}\right) e^{-\tau}$ is concave with respect to x, we can prove $f(x)$ to be concave by regarding it as the sum of infinite concave functions.

To prove that $g(x)$ is concave, we first expand the expectation

$$g(x) = -\frac{1}{\theta_k T}\ln \mathbb{E}\left\{e^{-\theta_k T B \log_2\left(1+\frac{h_k^m x}{N_0 B}\right)}\right\} = -\frac{1}{\theta_k T}\ln\left[\int_0^\infty e^{-\theta_k T B \log_2\left(1+\frac{\rho_k \tau x}{N_0 B}\right)} e^{-\tau} d\tau\right]$$

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$$= -\frac{1}{\theta_k T}\ln\left[\int_0^{\infty} e^{-\theta_k TB\log_2\left(1+\frac{\rho_k \tau x}{N_0 B}\right)-\tau} d\tau\right]. \quad (21)$$

Then we define a composite function

$$w(x) = u(v(x)) = u(v_1(x), v_2(x), \ldots, v_J(x)), \quad (22)$$

where

$$u(z) = -\frac{1}{\theta_k T}\ln\left[\sum\nolimits_{j=1}^{J} e^{z_j}\Delta\tau\right], \quad (23)$$

$$v_j(x) = -\theta_k TB\log_2\left(1+\frac{\rho_k j\Delta\tau x}{N_0 B}\right) - j\Delta\tau, \quad (24)$$

and $\Delta\tau \in R^+$. Since the LogSumExp function $\ln\left[\sum_{j=1}^{J} e^{z_j}\Delta\tau\right]$ is convex, multiplying it with a negative coefficient leads to the concavity of $u(z)$. It is also easy to prove that $v_j(x)$ is a convex function. According to the property of composite functions, $w(x)$ can be proved to be concave, because $u(z)$ is a monotonically non-decreasing concave function and $v_j(x)$ is a convex function. Finally, $g(x)$ is actually a special form of $w(x)$ as

$$g(x) = \lim_{\Delta\tau\to 0}\lim_{J\to\infty} w(x). \quad (25)$$

This proves the concavity of $g(x)$.

With the concavity of $f(x)$ and $g(x)$, we can continue the proof. Since h_k^m is i.i.d. with respect to m, we can prove that

$$\begin{aligned}\mathbb{E}\{C_k\} &= \mathbb{E}\left\{\sum_{m=1}^{N_k} B\log_2\left(1+\frac{P_k^m h_k^m}{N_0 B}\right)\right\}\\ &= \sum_{m=1}^{N_k}\mathbb{E}\left\{B\log_2\left(1+\frac{P_k^m h_k^m}{N_0 B}\right)\right\}\\ &= \sum_{m=1}^{N_k} f(P_k^m). \end{aligned} \quad (26)$$

According to the property of concave function $f(x)$, if $\sum_{m=1}^{N_k} P_k^m = P_{\rm sum}$ is given, $\mathbb{E}\{C_k\}$ is maximized when the power is equally allocated, i.e., $P_k^m = \frac{P_{\rm sum}}{N_k}$.

Afterward, the effective capacity of VUE k is rewritten as

$$\begin{aligned} EC_k(\theta_k) &= -\frac{1}{\theta_k T}\ln\mathbb{E}\left\{e^{-\theta_k TB\sum_{m=1}^{N_k}\log_2\left(1+\frac{P_k^m h_k^m}{N_0 B}\right)}\right\}\\ &= -\frac{1}{\theta_k T}\ln\mathbb{E}\left\{\prod_{m=1}^{N_k} e^{-\theta_k TB\log_2\left(1+\frac{P_k^m h_k^m}{N_0 B}\right)}\right\}\\ &= -\frac{1}{\theta_k T}\ln\prod_{m=1}^{N_k}\mathbb{E}\left\{e^{-\theta_k TB\log_2\left(1+\frac{P_k^m h_k^m}{N_0 B}\right)}\right\}\\ &= \sum_{m=1}^{N_k} -\frac{1}{\theta_k T}\ln\mathbb{E}\left\{e^{-\theta_k TB\log_2\left(1+\frac{P_k^m h_k^m}{N_0 B}\right)}\right\}\\ &= \sum_{m=1}^{N_k} g(P_k^m), \end{aligned} \quad (27)$$

where the third equation holds since h_k^m is i.i.d with respect to m. According to the property of the concave function $g(x)$, when θ_k is fixed and $\sum_{m=1}^{N_k} P_k^m = P_{\rm sum}$, the effective capacity $EC_k(\theta_k)$ is maximized when the power is equally allocated, i.e., $P_k^m = \frac{P_{\rm sum}}{N_k}$.

It is easy to prove that the constraint (8b) reaches the equality at the optimal point, thus the variable θ_k can be expressed as the function of P_k^m by substituting $P_{\rm D} = \sqrt{\frac{U\gamma}{K-1}}$ into (4), given by

$$\theta_k = -\frac{1}{\lambda_k T D_{\max}}\ln\frac{\sqrt{U\gamma}\mathbb{E}\{C_k\}}{\sqrt{K-1}\lambda_k}. \quad (28)$$

For $\sum_{m=1}^{N_k} P_k^m = P_{\rm sum}$, $\mathbb{E}\{C_k\}$ is maximized when $P_k^m = \frac{P_{\rm sum}}{N_k}$ as have been proved above, so θ_k is minimized. In addition, $EC_k(\theta_k)$ is monotonically decreasing with θ_k, which has been proved in [33]. If $P_{\rm sum}$ is fixed, θ_k is minimized when $P_k^m = \frac{P_{\rm sum}}{N_k}$. If $P_{\rm sum}$ and θ_k are fixed, $EC_k(\theta_k)$ is maximized when $P_k^m = \frac{P_{\rm sum}}{N_k}$ as has been proved above. Therefore, if $P_{\rm sum}$ is fixed, it is clear that $EC_k(\theta_k)$ is maximized when $P_k^m = \frac{P_{\rm sum}}{N_k}$. On the contrary, when $EC_k(\theta_k) = \lambda_k$, the sum power $P_{\rm sum}$ is minimized when $P_k^m = \frac{P_{\rm sum}}{N_k}$. This ends the proof.

APPENDIX B
OPTIMALITY OF ALGORITHM 1

When U is fixed, the optimal value of the objective function in problem $\mathcal{P}2_k$ can be expressed as the function of N_k, denoted by $Y_k(N_k)$. The optimal solution to $\mathcal{P}2_k$ with $N_k = n$ can be regarded as a feasible but not optimal solution to $\mathcal{P}2_k$ with $N_k = n+1$ and $Y_k(n) > Y_k(n+1)$, since the former always allocates zero power to the $(n+1)$-th RB. Therefore, $Y_k(N_k)$ is a monotonically decreasing function of N_k.

When applying Algorithm 1, the index of the VUE with the maximum sum power among all VUEs is given as

$$i_z = \underset{k}{\rm argmax}\, Y_k(N_k^{z-1}), \quad (29)$$

where N_k^z is the number of RBs allocated to VUE k at the end of the z-th round of Lines 3–6 and $\sum_{k=1}^{K} N_k^z = z$. Therefore, the update of N_k^z can be expressed as

$$\begin{aligned} N_{i_z}^z &= N_{i_z}^{z-1} + 1,\\ N_k^z &= N_k^{z-1}, \quad \forall k \neq i_z. \end{aligned} \quad (30)$$

To prove the optimality of algorithm 1, we define $Y_k(0) = \infty$ and start from the K-th round where $N_k^K = 1, \forall k \in \mathcal{K}$. Obviously, $N_k = 1, \forall k \in \mathcal{K}$ is the optimal solution when $\sum_{k=1}^{K} N_k = K$, since any other allocation result will lead to the infinite value of the objective function (8). Next, we will prove that

$$Y_k(N_k^z - 1) \geqslant Y_{i_z}(N_{i_z}^z - 1), \quad \forall k \in \mathcal{K}, z \geqslant K. \quad (31)$$

Since the inequality holds when $z = K$, we only need to prove that $Y_k(N_k^{z+1} - 1) \geqslant Y_{i_{z+1}}(N_{i_{z+1}}^{z+1} - 1)$ when $Y_k(N_k^z - 1) \geqslant Y_{i_z}(N_{i_z}^z - 1)$, as

$$\begin{aligned} Y_k(N_k^{z+1} - 1) &\geqslant Y_{i_{z+1}}(N_{i_{z+1}}^{z+1} - 1), \quad k = i_{z+1},\\ Y_k(N_k^{z+1} - 1) &\overset{\text{a}}{=} Y_k(N_k^z - 1)\\ &\overset{\text{b}}{\geqslant} Y_{i_z}(N_{i_z}^z - 1)\\ &\overset{\text{c}}{=} Y_{i_z}(N_{i_z}^{z-1})\\ &\overset{\text{d}}{\geqslant} Y_{i_{z+1}}(N_{i_{z+1}}^{z-1})\\ &\overset{\text{e}}{\geqslant} Y_{i_{z+1}}(N_{i_{z+1}}^{z})\\ &\overset{\text{f}}{=} Y_{i_{z+1}}(N_{i_{z+1}}^{z+1} - 1), \quad k \neq i_{z+1}, \forall k \in \mathcal{K}, \end{aligned} \quad (32)$$

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where ⓐ, ⓒ and ⓕ follow from (30), ⓑ follows from (31), ⓓ follows from (29), ⓔ follows from (30) and the monotonically decreasing property of $Y_k(N_k)$. Therefore, property (31) can be proved through mathematical induction.

Now we prove that N_k^{M-U} is the optimal solution to problem $\mathcal{P}1$ when U is fixed. Let $\hat{N}_k$ denote an arbitrary solution which satisfies the constraint $\sum_{k=1}^{K}\hat{N}_k = M-U$ and is different from N_k^{M-U}, that is, $\hat{N}_k \neq N_k^{M-U}, \exists k \in \mathcal{K}$. Then the following inequality holds

$$\hat{N}_k < N_k^{M-U}, \quad \exists k \in \mathcal{K}, \tag{33}$$

which can be proved by using reduction to absurdity as follows. If $\hat{N}_k \geqslant N_k^{M-U}, \forall k \in \mathcal{K}$, then $\sum_{k=1}^{K}\hat{N}_k = M-U \geqslant \sum_{k=1}^{K} N_k^{M-U} = M-U$ which leads to $\sum_{k=1}^{K}\hat{N}_k = \sum_{k=1}^{K} N_k^{M-U}$. The equation holds only when $\hat{N}_k = N_k^{M-U}, \forall k \in \mathcal{K}$, which conflicts with the assumption that $\hat{N}_k \neq N_k^{M-U}, \exists k \in \mathcal{K}$. Without loss of generality, we assume that $\hat{N}_l < N_l^{M-U}$ which leads to

$$\begin{aligned}\max_k\left(Y_k(\hat{N}_k)\right) &\geqslant Y_l(\hat{N}_l)\\ &\geqslant Y_l(N_l^{M-U}-1)\\ &\geqslant Y_{i_{M-U}}(N_{i_{M-U}}^{M-U}-1)\\ &= Y_{i_{M-U}}(N_{i_{M-U}}^{M-U-1})\\ &= \max_k\left(Y_k(N_k^{M-U-1})\right)\\ &\geqslant \max_k\left(Y_k(N_k^{M-U})\right).\end{aligned} \tag{34}$$

Therefore, it is proved that $\max_k\left(Y_k(\hat{N}_k)\right) \geqslant \max_k\left(Y_k(N_k^{M-U})\right)$ for arbitrary $\hat{N}_k$, which implies that N_k^{M-U} is the optimal solution to problem $\mathcal{P}1$ when U is fixed. This ends the proof.

APPENDIX C
PROOF OF THEOREM 3

Unfolding the objective function of $\mathcal{P}3$ results in

$$\begin{aligned}L(\vec{p}_k(n)) = &\sum_{m=1}^{M_k(n)} p_k^m(n) - \alpha w T B \sum_{m=1}^{M_k(n)} \log_2\left(1+\frac{p_k^m(n)}{\sigma_k^m(n)}\right)\\ &-\alpha\sum_{i=w+1}^{D_{\max}} iQ_k^i(n) + \alpha I\sum_{i=w+1}^{D_{\max}} Q_k^i(n),\end{aligned} \tag{35}$$

where $\vec{p}_k(n)$ denotes the set $\{p_k(n), k=1,\ldots,M_k(n)\}$ and $\sigma_k^m(n) = \frac{N_0 B}{h_k^m(n)}$ is the equivalent channel noise. The third and fourth terms of $L(\vec{p}_k(n))$ are constant and can be omitted.

Denote a feasible solution to $\mathcal{P}3$ as $\bar{\vec{p}}_k(n) = \left\{\bar{p}_k^1(n),\ldots,\bar{p}_k^{M_k(n)}(n)\right\}$ and calculate $\bar{C}_k(n) = B\sum_{m=1}^{M_k(n)}\log_2\left(1+\frac{\bar{p}_k^m(n)}{\sigma_k^m(n)}\right)$. Next, we find the water pouring solution which makes the channel capacity equal to $\bar{C}_k(n)$. That is, $\hat{p}_k^m(n) = \max\{\eta-\sigma_k^m(n),0\}$ and $B\sum_{m=1}^{M_k(n)}\log_2\left(1+\frac{\hat{p}_k^m(n)}{\sigma_k^m(n)}\right) = \bar{C}_k(n)$. According to the property of the water pouring solution, $\sum_{m=1}^{M_k(n)}\hat{p}_k^m(n) \leqslant \sum_{m=1}^{M_k(n)}\bar{p}_k^m(n)$, the equality holds when $\hat{p}_k^m(n) = \bar{p}_k^m(n)$. Therefore, $L(\hat{\vec{p}}_k(n)) \leqslant L(\bar{\vec{p}}_k(n))$ since the first item is smaller and the second item is the same. Furthermore, $\hat{\vec{p}}_k(n)$ also satisfies constraints (18b) and (18c) since the channel capacity is the same and the sum power is smaller compared with $\bar{\vec{p}}_k(n)$. This ends the proof.

APPENDIX D
PROOF OF THEOREM 4

Denote $\eta(x)$ as the water pouring threshold when the sum power $\sum_{m=1}^{M_k(n)} p_k^m(n) = x$. The first-order derivative of $H(x)$ is then expressed as

$$\begin{aligned}H'(x) &= 1-\alpha J'\left(G_k(x)\right)\left(G_k'(x)\right)\\ &= 1-\frac{\alpha w(x) T B}{\eta(x)\ln 2},\end{aligned} \tag{36}$$

where $w(x)$ denotes the maximum queuing latency of the data in the queue after transmission when $\sum_{m=1}^{M_k(n)} p_k^m(n) = x$. Since $w(x)$ is a monotonically non-increasing function of x which is always non-negative and $\eta(x)$ is a monotonically increasing function of x which is always positive, it is proved that $H'(x)$ is a monotonically increasing function of x. Therefore, $H(x)$ is a convex function, which ends the proof.

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